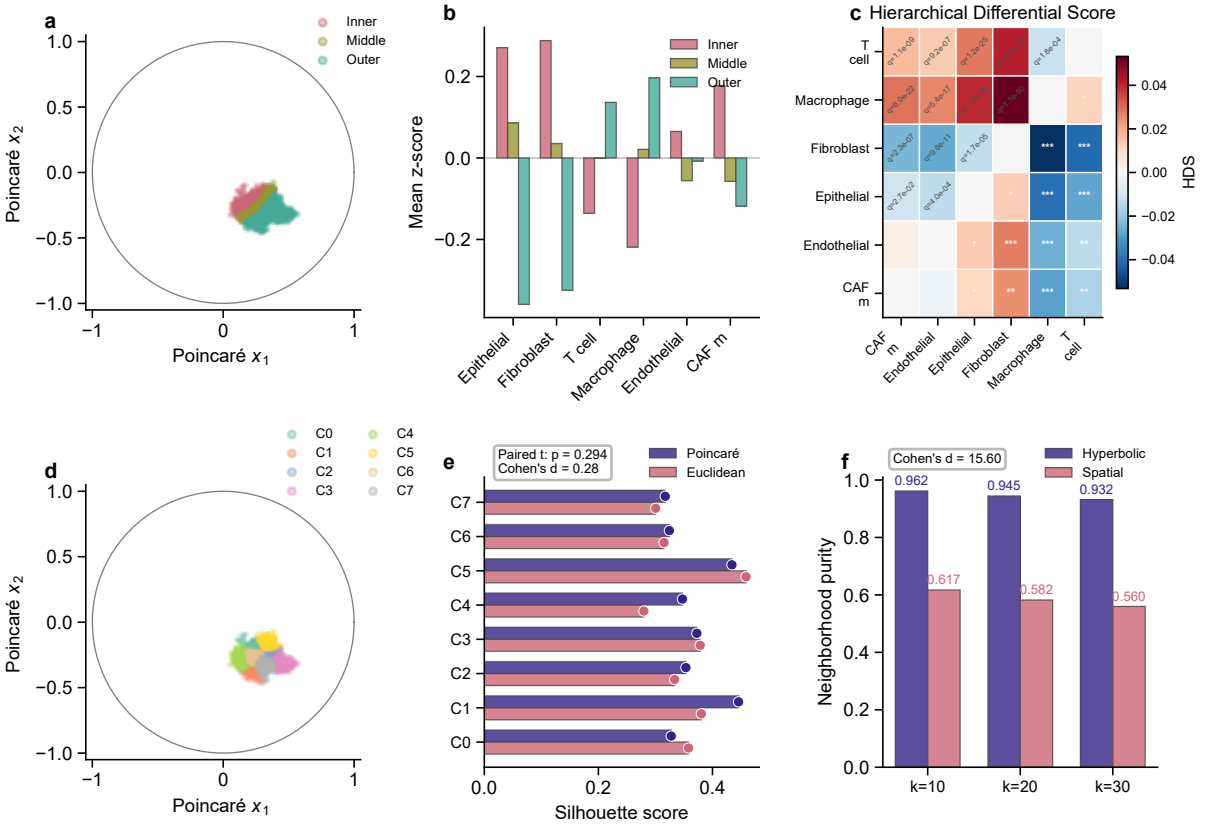
a
Inner
Middle
Outer
Poincaré x_1
Poincaré x_2
b
Mean z-score
Epithelial
Fibroblast
T cell
Macrophage
Endothelial
CAF m
c Hierarchical Differential Score
HDS
d
C0
C1
C2
C3
C4
C5
C6
C7
e
Paired t: p = 0.294
Cohen's d = 0.28
Poincaré
Euclidean
Silhouette score
f
Cohen's d = 15.60
Hyperbolic
Spatial
Neighborhood purity
0.962
0.617
0.945
0.582
0.932
0.560
k=10
k=20
k=30